

SHAWMICA SIVATHARAN

AI & Machine Learning Engineer Intern

Software Engineer Intern

Vaddukoddai, Jaffna, Sri Lanka, 40000

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Profile Summary

Passionate Information Technology undergraduate at Rajarata University of Sri Lanka and aspiring Software Engineer Intern with a strong focus on Machine Learning. Proficient in Python, TensorFlow, PyTorch, and OpenCV, with hands-on experience integrating AI/ML models into real-world web and vision-based applications. Skilled in full-stack development (React, Django) and driven by curiosity to build data-driven, scalable, and human-centered AI solutions. Dedicated to crafting scalable, intuitive, and visually engaging web applications that deliver real-world impact with the problem-solving mindset. Eager to contribute to innovative projects, collaborate with industry experts, and grow as a versatile ML engineer.

Education

BSc in Information Technology

2023 – Present

Rajarata University of Sri Lanka

CGPA - 3.75/4

Secondary Education

2006 – 2020

Jaffna College, Jaffna, Sri Lanka

Physical Science Stream (Advanced Level)

Mathematics-B, Chemistry-C, Physics-C

Technical Skills

- **Programming Languages:** Python, Java, C, C++, JavaScript, SQL, Prolog
- **AI & Machine Learning:** TensorFlow, Keras, PyTorch, scikit-learn, OpenCV, Tesseract OCR, Hugging Face Transformers, RAG (Retrieval-Augmented Generation), MCP (Model Context Protocol)
- **Computer Vision:** OpenCV, cvzone, Pycaw
- **Data Analysis & Visualization:** Pandas, Numpy, Matplotlib
- **Frontend:** React.js, Bootstrap, Tailwind CSS
- **Backend:** Django, PHP, Node.js
- **Databases:** MySQL, MongoDB, SQLite
- **Mobile Development:** Kotlin, React Native, Java
- **Cloud & Development Tools:** Vercel, AWS, Docker, Firebase Hosting
- **Development Tools:** Postman, Git, GitHub, Jupyter Notebook, Figma, Canva
- **Other:** Data Structures, Algorithms, OOP
- **Soft Skills:** Problem-Solving, Communication, Teamwork, Leadership, Adaptability, Project Management

Projects

Student Grade Prediction Model | Artificial Neural Networks (ANN)

Jun 2025 – Aug 2025

- Led the development of an ANN-based model that predicts final student grades (A, B, C, F) using engineered features. Designed and optimized the neural network architecture while overseeing data preprocessing, feature engineering, and model evaluation as team leader. Built a system where students input study hours, credits, quiz scores, attendance, engagement, and missed deadlines, which are transformed into meaningful predictors for accurate grade classification.
- **Technologies Used:** Python, NumPy, Pandas, Scikit-learn, TensorFlow/Keras.

Fungal Colony Analysis on Coconut Flesh | Image-Based Growth Study

Mar 2024 – Aug 2024

- Performed a 14-day image processing study to quantify black fungal growth on coconut flesh using controlled imaging and preprocessing techniques. Applied segmentation methods such as dilation, adaptive thresholding, and floodfill to isolate fungal regions and measure growth patterns.
- **Technologies Used:** Python, OpenCV, Matplotlib.

Mathly | AI-Powered Multilingual Math Solver

Jul 2025 - Aug 2025

- Developed an AI tool that uses OCR to read printed equations, solves them with SymPy, and provides step-by-step explanations with multilingual voice output.
- **Technologies Used:** Python, OpenCV, Tesseract OCR, SymPy, gTTS.

- Built a hand-gesture-based presentation controller enabling slide navigation, annotations, erasing, and on-screen highlighting using OpenCV and cvzone's HandTrackingModule.
- **Technologies Used:** Python, OpenCV, cvzone.

Hand Gesture Volume Control | Computer Vision Interaction System

Jul 2024

- Built a real-time hand gesture-based volume controller using OpenCV for hand tracking and Pycaw for system-level audio control, enabling volume adjustment through finger-pinch distance.
- **Technologies Used:** Python, OpenCV, Pycaw.

Gesture-Controlled Presentation Tool | Computer Vision Slide Navigator

Jul 2024

- Built a real-time hand gesture-based slide presentation controller enabling slide navigation, drawing, annotation undo, on-screen highlighting, and picture-in-picture presenter view using OpenCV and cvzone's HandTrackingModule.
- **Technologies Used:** Python, OpenCV, cvzone.

Real-Time AI Assistant Using RAG, LangChain & ChromaDB | AI-Powered Document Q&A System

May 2026

- Built a real-time AI assistant that allows users to upload PDFs and interact with documents through natural language conversations using a RAG pipeline powered by HuggingFace embeddings, ChromaDB vector storage, semantic search, Groq Llama 3.3, FastAPI, and Streamlit.
- **Technologies Used:** Python, LangChain, ChromaDB, FastAPI, Streamlit, HuggingFace, Groq, PyPDF.

ClassPulse: AI-Powered Student Engagement Analytics Platform | Final Year Research Project

Feb 2025 – March 2026

- Developed an AI-powered online learning platform that enhances student engagement in live video conferencing sessions through predictive learner clustering, personalized feedback generation, targeted quiz interactions, real-time analytics, and network-aware engagement monitoring.
- **Technologies Used:** React, TypeScript, Python, FastAPI, MongoDB, MySQL, TensorFlow, Scikit-learn, WebSocket, WebRTC, Zoom API, Tailwind CSS.

Certifications

- AI/ML Engineer - Stage 1 – CODL (SLIIT), 2025
- AI/ML Engineer - Stage 2 – CODL (SLIIT), 2025
- Machine Learning, Artificial Intelligence – Simplilearn, 2025
- Python Django 101 - Simplilearn, 2025
- Exploratory Data Analysis for ML – IBM, 2025
- Foundations of Project Management - CODL (UoM), 2024
- IEEE Xtreme 18.0 – Rank 72(885), 2024
- RUSL Xtreme 1.0 – First Place, 2024
- Xtreme Encode, IEEE Xtreme 18.0 Region 10, 2024
- INSL(North Central Province) – First Place, 2025
- Problem Solving (Basic) - Hacker Rank, 2024
- Java Programming - DMI Education, 2022
- Python Programming - CODL (UoM), 2023
- Front-End Web Development - CODL (UoM), 2023
- Introduction to Generative AI – AWS, 2025
- Microsoft Azure - Microsoft Learn Ambassadors, 2024
- API Beginner Path, API Testing Path - Postman Academy, 2025
- Agile Project Management - CODL (UoM), 2024
- Prolog Programming Course - Udemy, 2025
- NLP and Text Mining - Simplilearn, 2025

Club and Societies

- Publicity Team Member - IEEE Student Branch of RUSL
- Student Member - IEEE, IEEE Computer Society, IEEE Women in Engineering
- Board of Prefects, Jaffna College

Extra-Curricular Activities

- Grade 3 Music Theory - The Associated Board of the Royal Schools of Music
- Grade 3 Piano - The Associated Board of the Royal Schools of Music

References

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